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Division: A2

ASSIGNMENT NO: 5

TITLE: Develop a Java application demonstrating the use of ArrayList, Vector, and StringBuffer for dynamic data storage, management, and string manipulation.

Problem Statement: Develop a Java application demonstrating the use of ArrayList, Vector, and StringBuffer for dynamic data storage, management, and string manipulation.

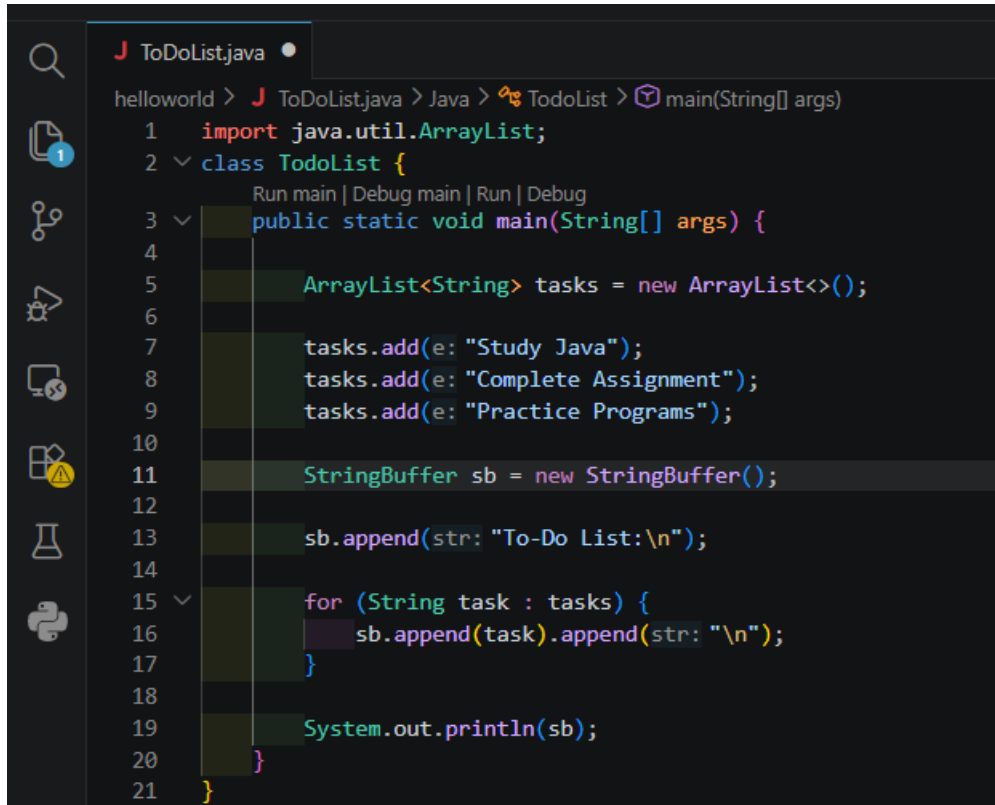
Learning Objectives: To implement dynamic data storage and string manipulation using ArrayList, Vector, and StringBuffer.

Theory: ArrayList and Vector are dynamic array implementations provided by the Java Collections Framework. ArrayList offers faster performance in single-threaded applications, whereas Vector provides synchronized operations for thread safety. StringBuffer is a mutable sequence of characters that allows efficient string modification without creating new objects. These classes are commonly used in applications requiring dynamic memory allocation and efficient string processing.

Exercise:

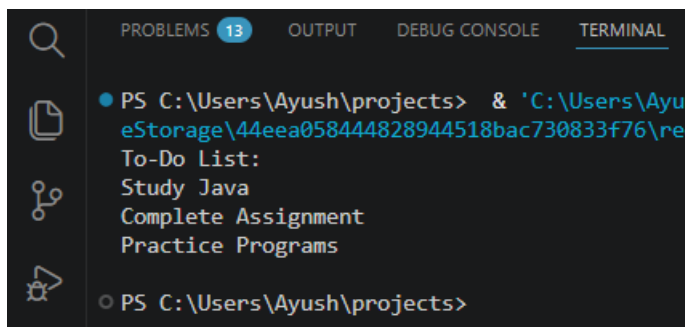
1. Create a To-Do List application using ArrayList to store tasks and StringBuffer to display tasks.

Program:



```
helloworld > J ToDoList.java > Java > TodoList > main(String[] args)
1  import java.util.ArrayList;
2  class TodoList {
3      public static void main(String[] args) {
4
5          ArrayList<String> tasks = new ArrayList<>();
6
7          tasks.add(e: "Study Java");
8          tasks.add(e: "Complete Assignment");
9          tasks.add(e: "Practice Programs");
10
11         StringBuffer sb = new StringBuffer();
12
13         sb.append(str: "To-Do List:\n");
14
15         for (String task : tasks) {
16             sb.append(task).append(str: "\n");
17         }
18
19         System.out.println(sb);
20     }
21 }
```

Output:



```
PROBLEMS 13 OUTPUT DEBUG CONSOLE TERMINAL
● PS C:\Users\Ayush\projects> & 'C:\Users\Ayush\AppData\Local\Microsoft\Windows\Temporary Internet Files\Content.IE5\XV1QJQJQ\re
To-Do List:
Study Java
Complete Assignment
Practice Programs
○ PS C:\Users\Ayush\projects>
```

2. Create a Student Course Registration System using ArrayList to store the list of courses registered by a student and StringBuffer to generate and display the registered course list. The application should allow users to add, remove, and view registered courses.

Program:

```
helloworld > J CourseRegistration.java > ...
1  import java.util.ArrayList;
2  class CourseRegistration {
3      Run main | Debug main | Run | Debug
4      public static void main(String[] args) {
5          ArrayList<String> courses = new ArrayList<>();
6
7          courses.add(e: "Java");
8          courses.add(e: "Python");
9          courses.add(e: "C++");
10
11         courses.remove(o: "Python");
12
13         StringBuffer sb = new StringBuffer();
14         sb.append(str: "Registered Courses:\n");
15
16         for (String course : courses) {
17             sb.append(course).append(str: "\n");
18         }
19
20         System.out.println(sb);
21     }
22 }
```

Output:

```
PROBLEMS 12 OUTPUT DEBUG CONSOLE
• PS C:\Users\Ayush\projects> &
eStorage\44eea058444828944518ba
Registered Courses:
Java
C++
○ PS C:\Users\Ayush\projects>
```